Experiment 2 Pre-Lab Report

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1 Task 1: Summary of Theoretical Concepts

RF receiver architectures are bounded at lower signal levels by thermal and internal component noise, and at upper levels by non-linear gain saturation and intermodulation distortion.

1.1 Noise Figure (NF) and Noise Temperature (T_e)

The Noise Factor (F) and Noise Figure (NF) measure the degradation of the Signal-to-Noise Ratio (SNR) as a signal propagates through a receiver processing chain. Thermal noise power over bandwidth B at reference temperature $T_0 = 290$ K is given by $N_{\text{in}} = kT_0B$ (where $kT_0 = -174$ dBm/Hz). An active or lossy stage adds equivalent input noise temperature T_e:

$$F = \frac{SNR_{\text{in}}}{SNR_{\text{out}}} = 1 + \frac{T_e}{T_0}, \quad NF = 10\log_{10}(F) \tag{1}$$

The input-referred noise floor (N_{floor}) over bandwidth B is expressed as:

$$N_{\text{floor}} = -174 \text{ dBm/Hz} + NF + 10\log_{10}(B) \tag{2}$$

1.2 Y-Factor Noise Measurement Method

The Y-Factor method utilizes a calibrated noise source operating in two states: OFF ($T_c \approx T_0$) and ON ($T_h \gg T_0$), defined by the Excess Noise Ratio (ENR):

$$ENR = \frac{T_h - T_0}{T_0}, \quad ENR_{\text{dB}} = 10\log_{10}(ENR) \tag{3}$$

Measuring the ratio of output noise powers $Y = N_{\text{on}}/N_{\text{off}}$ determines the Noise Factor without needing absolute power calibration, gain knowledge, or noise bandwidth:

$$Y = \frac{T_h + T_e}{T_0 + T_e} \implies F = \frac{ENR}{Y - 1} \tag{4}$$

1.3 Third-Order Intercept Point (IIP_3) and Gain Compression ($P_{1\text{dB}}$)

Under two-tone excitation at f_1 and f_2, 3rd-order non-linearities generate intermodulation products (IM3) at $2f_1 - f_2$ and $2f_2 - f_1$. While fundamental tones increase at a 1 dB/dB rate, IM3 products grow at 3 dB/dB. The theoretical power level where extended fundamental and IM3 powers intersect defines the Input Third-Order Intercept Point (IIP_3).

2 Task 2: Analytical Solution of Theoretical Questions

2.1 Question 1: Noise Temperature and Friis Formula

(a) Equivalent Noise Temperature Calculation ($T_0 = 290$ K): Using $T_e = (10^{NF/10} - 1)T_0$:

- $NF = 1$ dB $\implies F = 1.2589 \implies T_e = (1.2589 - 1) \times 290 = 75.1$ K
- $NF = 3$ dB $\implies F = 1.9953 \implies T_e = (1.9953 - 1) \times 290 = 288.6$ K
- $NF = 6$ dB $\implies F = 3.9811 \implies T_e = (3.9811 - 1) \times 290 = 864.5$ K
- $NF = 10$ dB $\implies F = 10.0000 \implies T_e = (10 - 1) \times 290 = 2610.0$ K

(b) Total Noise Figure of 3-Stage Chain: Original chain configuration: Stage 1 = LNA ($G_1 = 20$ dB $= 100, F_1 = 1.2589$), Stage 2 = Filter ($G_2 = -3$ dB $= 0.5012, F_2 = 1.9953$), Stage 3 = Mixer/ADC ($F_3 = 31.6228$). Applying Friis formula:

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} = 1.2589 + \frac{0.9953}{100} + \frac{30.6228}{50.12} = 1.87985 \implies NF_{\text{total}} \approx 2.74 \text{ dB} \quad (5)$$

(c) Filter Relocated Before LNA: Moving the filter to the frontend alters the chain structure to: Stage 1 = Filter ($G_1 = 0.5012, F_1 = 1.9953$), Stage 2 = LNA ($G_2 = 100, F_2 = 1.2589$), Stage 3 = Mixer/ADC ($F_3 = 31.6228$). Recalculating Friis formula:

$$F_{\text{new}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} = 1.9953 + \frac{1.2589 - 1}{0.5012} + \frac{31.6228 - 1}{0.5012 \times 100} = 3.1229 \quad (6)$$

$$NF_{\text{new}} = 10 \log_{10}(3.1229) \approx 4.95 \text{ dB} \quad (7)$$

2.2 Question 2: Y-Factor Derivation and Attenuation

(a) Derivation for $T_c \neq T_0$: Starting from $Y = \frac{T_h + T_e}{T_c + T_e} \implies T_e(Y - 1) = T_h - YT_c$. Substituting $T_e = T_0(F - 1)$ and $T_h = T_0(1 + ENR)$:

$$T_0(F-1)(Y-1) = T_0(1+ENR) - YT_c \implies (F-1)(Y-1) = ENR - Y\left(\frac{T_c}{T_0} - 1\right) - (Y-1) \quad (8)$$

Dividing by $(Y - 1)$ and adding 1 yields the exact general relation:

$$F = \frac{ENR - Y\left(\frac{T_c}{T_0} - 1\right)}{Y - 1} \quad (9)$$

When $T_c = T_0$, the second numerator term vanishes, simplifying to $F = \frac{ENR}{Y-1}$.

(b) Attenuator A at Temperature T_0: An attenuator reduces effective ENR to $ENR_{\text{eff}} = \frac{ENR}{A} \implies ENR_{\text{eff, dB}} = ENR_{\text{dB}} - A_{\text{dB}}$.

2.3 Question 3: Error Sensitivity in Y-Factor Method

For a readout error $\Delta Y = 0.1$ dB $\implies Y' = 1.0233 \times Y$:

- **Case 1 ($NF = 6$ dB, $ENR = 15$ dB):** True $F = 3.9811 \implies Y = 8.9433$. With +0.1 dB error, $F' = 3.8793 \implies NF' = 5.89$ dB ($\Delta NF = -0.11$ dB).
- **Case 2 ($NF = 25$ dB, $ENR = 15$ dB):** True $F = 316.2278 \implies Y = 1.1000$. With +0.1 dB error, $F' = 251.73 \implies NF' = 24.01$ dB ($\Delta NF = -0.99$ dB).

Rule: Select ENR high enough relative to DUT noise figure such that $Y \geq 3$ dB to avoid error magnification in $(Y-1)$ denominator.

2.4 Question 4: Non-linearity and IIP3

For system $y = a_1 x + a_3 x^3$ under two tones, $A_{\text{IM3}} = \frac{3}{4}|a_3|S^3 \implies IIP_3 = P_{\text{in}} + \frac{\Delta}{2}$. Under pure 3rd-order memoryless assumptions, $IIP_3 \approx P_{\text{1dB}} + 9.6$ dB.

Limitations: (1) Ignores higher-order non-linear terms and memory effects; (2) ADC hard clipping violates the 3rd-order polynomial slope.

2.5 Question 5: Spurious-Free Dynamic Range (SFDR)

Given $NF = 8$ dB, $IIP_3 = -10$ dBm:

- $BW = 10$ kHz $\implies N_{\text{floor}} = -126$ dBm $\implies SFDR = \frac{2}{3}(-10 - (-126)) = 77.33$ dB.
- $BW = 200$ kHz $\implies N_{\text{floor}} = -112.99$ dBm $\implies SFDR = \frac{2}{3}(-10 - (-112.99)) = 68.66$ dB.

SFDR depends on bandwidth because integrated thermal noise floor scales with bandwidth (kT_0B).

2.6 Question 6: SDR Receiver Gain Trade-offs

Increasing receiver gain by +10 dB lowers input-referred ADC noise floor by 10 dB while lowering input P_{1dB} by 10 dB. Gain increases stop improving NF once LNA gain renders ADC noise contribution negligible.

3 Task 3: Measurement Design Calculations (Table 4)

Table 1: Design Calculations for Measurement Plan (Task 3)

Parameter	**Value**	**Numerical Rationale**
Suitable ENR for $NF \approx 7$ dB and $NF \approx 30$ dB	15 dB / 35 dB	Ensures $Y \geq 3$ dB to avoid readout error amplification.
Required Attenuation after Noise Source	10 dB attenuator	Prevents receiver saturation during the ON state.
Averaging Time for < 0.05 dB fluctuation	$T_{\text{avg}} \geq 25$ ms	Reduces statistical noise variance in 400 kHz band.
Single-Tone Sweep Range & Step	-60 to -10 dBm (1 dB step)	Pinpoints P_{1dB} without exceeding max safe input limit.
Two-Tone Frequencies & Spacing	$f_1 = 300$ kHz, $f_2 = 400$ kHz	Places IM3 products (200 kHz, 500 kHz) inside analysis band.
Tone Power for IM3 $\geq$ 10 dB above noise	$P_{\text{tone}} \approx -35$ dBm	Ensures clear spectral resolution of IM3 products.

4 Task 4: Offline Receiver Model Simulation (GNU Radio)

4.1 Y-Factor Noise Figure Measurement

Using the `lab02_sim.grc` flowgraph across two gain settings:

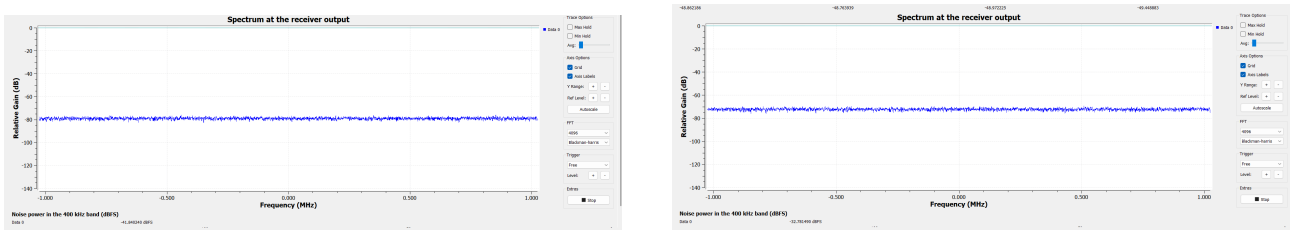
- **Receiver Gain = 40.0 dB (ENR = 15 dB):** Figure 1 displays the spectra. Measured $N_{\text{off}} = -41.85$ dBFS and $N_{\text{on}} = -32.78$ dBFS.

$$Y_{\text{dB}} = -32.78 - (-41.85) = 9.07 \text{ dB} \implies Y = 8.07 \implies NF = 10\log_{10}\left(\frac{31.6228}{7.07}\right) \approx 6.50 \text{ dB} \tag{10}$$

- **Receiver Gain = 10.0 dB (ENR = 15 dB):** Measured $N_{\text{off}} = -51.40$ dBFS and $N_{\text{on}} = -51.10$ dBFS.

$$Y_{\text{dB}} = 0.30 \text{ dB} \implies Y = 1.0715 \implies NF = 10\log_{10}\left(\frac{31.6228}{0.0715}\right) \approx 26.46 \text{ dB} \tag{11}$$

Design Evaluation: At 10.0 dB gain, using $ENR = 15$ dB results in $Y = 0.30$ dB < 3 dB, introducing severe sensitivity to readout noise (as derived in Task 2.3). In practice, an $ENR \approx 35$ dB source must be selected for low-gain states to maintain $Y \geq 3$ dB.



(a) OFF State ($N_{\text{off}} = -41.85$ dBFS) (b) ON State ($N_{\text{on}} = -32.78$ dBFS)

Figure 1: Output noise spectrum in OFF and ON states at 40.0 dB receiver gain.

4.2 Single-Tone Compression (P_{1dB}) and Two-Tone Intermodulation (IIP_3)

- **Single-Tone Compression at Gain = 0.0 dB:** Sweeping input power shows that at $P_{\text{in}} = -23.300$ dBm, fundamental output tone f_1 reaches $+0.2406$ dBFS (Figure 2), reaching ADC full-scale saturation ($P_{\text{1dB, in}} \approx -23.3$ dBm).

- **Two-Tone Intermodulation at Gain = 0.0 dB ($P_{\text{in}} = -30.0$ dBm):** Figure 3a shows fundamental tones $f_1, f_2 = -10.00$ dBFS and 3rd-order products $(2f_1 - f_2, 2f_2 - f_1) = -25.79$ dBFS.

$$\Delta = P_{\text{fund}} - P_{\text{IM3}} = -10.00 - (-25.79) = +15.79 \text{ dB} \tag{12}$$

$$IIP_3 = P_{\text{in}} + \frac{\Delta}{2} = -30.0 + \frac{15.79}{2} = -22.10 \text{ dBm} \tag{13}$$

Comparing with $P_{\text{1dB}} + 9.6$ dB $= -23.3 + 9.6 = -13.7$ dBm, the simulated IIP_3 (-22.10 dBm) is lower due to non-polynomial ADC quantization saturation effects.

- **Two-Tone Hard Clipping at Gain = 40.0 dB ($P_{\text{in}} = -30.0$ dBm):** Figure 3b displays total output reaching $+2.86$ dBFS, triggering severe ADC hard clipping and rendering standard linear IIP_3 extrapolation invalid.

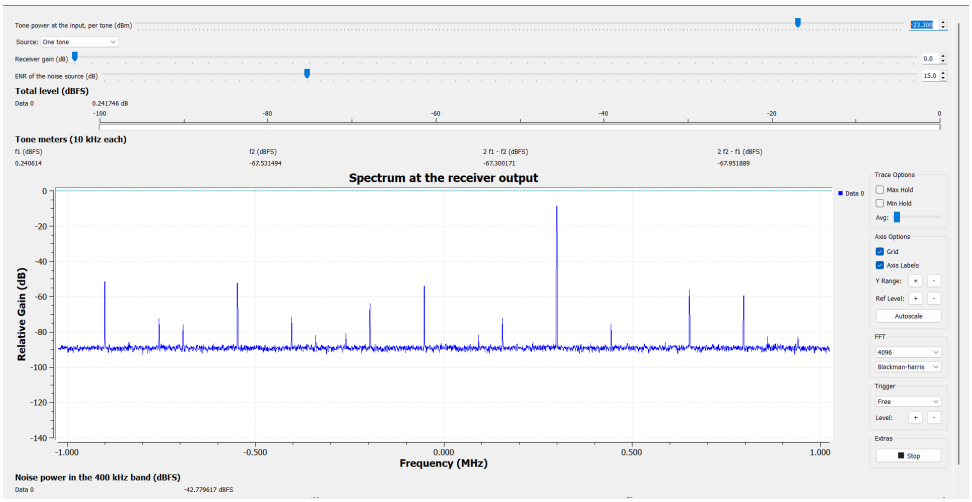
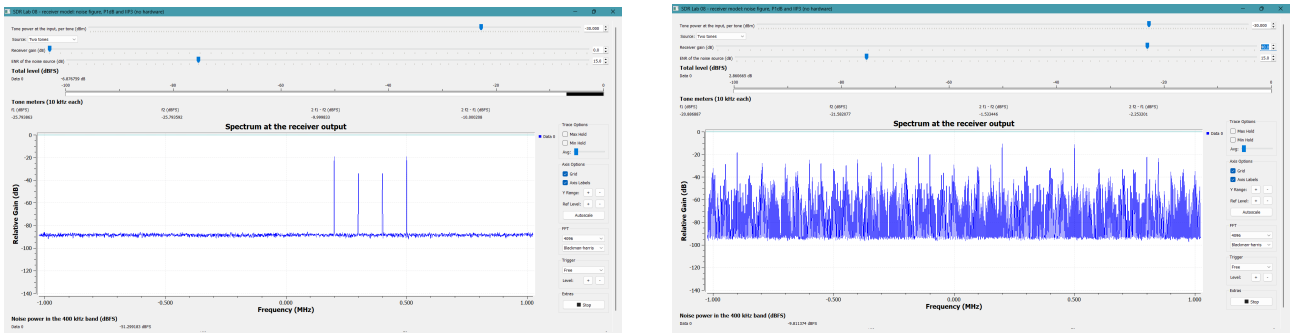


Figure 2: Single-tone output spectrum at gain compression threshold ($P_{\text{in}} = -23.3$ dBm, $G = 0.0$ dB).



(a) Gain = 0.0 dB (Clean IM3, $IIP_3 = -22.1$ dBm)

(b) Gain = 40.0 dB (ADC Hard Clipping)

Figure 3: Two-tone intermodulation spectrum comparison under linear vs. saturated gain conditions.

Table 2: Receiver Dynamic Performance across Gain Settings

Receiver Gain (dB)	**Noise Figure (NF, dB)**	**Input $P_{1\text{dB}}$ (dBm)**
0.0	> 30.0	-23.3
10.0	26.46	-33.3
40.0	6.50	-63.3

5 Task 5: Offline Antenna Frequency Response Simulation

5.1 Commercial Specifications (Table 1)

Operational frequency bands from manufacturer specs: $700 - 960$ MHz and $1710 - 2700$ MHz.

5.2 Simulated 3-dB Bandwidth Measurement (Table 5)

Executing `lab02_antenna_response_simulation.py` generates the normalized transmission curve shown in Figure 4. Lower cutoff $f_L = 800.0$ MHz and upper cutoff $f_H = 1800.0$ MHz are measured at the -3 dB attenuation points relative to passband peak.

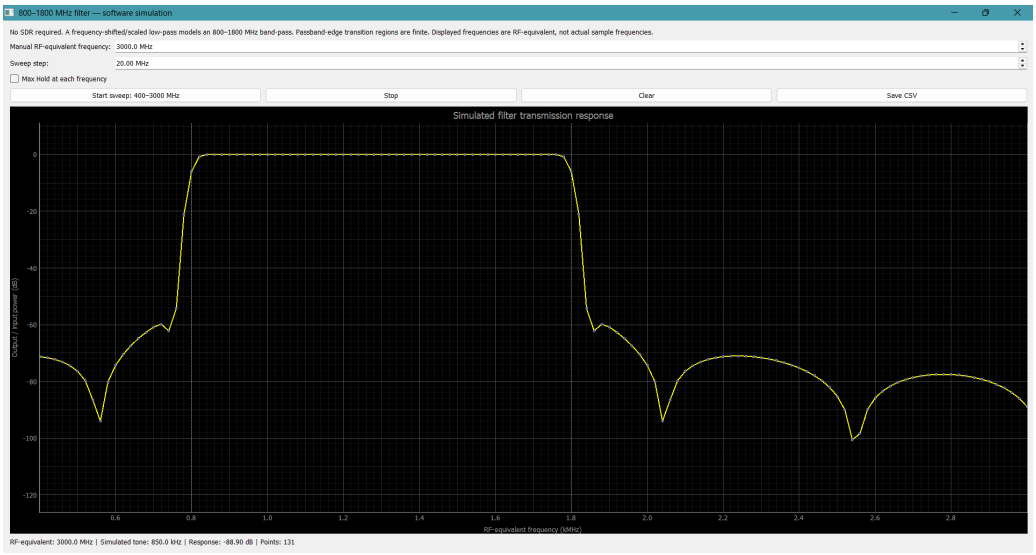


Figure 4: Simulated antenna transmission response displaying 800–1800 MHz 3-dB passband region.

Table 3: Cutoff Frequencies and 3-dB Bandwidth (Table 5)

Parameter	**Value (MHz)**
Lower Cutoff Frequency (f_L)	800.0 MHz
Upper Cutoff Frequency (f_H)	1800.0 MHz
3-dB Bandwidth ($BW = f_H - f_L$)	1000.0 MHz

6 Task 6: Pre-Laboratory Performance Predictions for Hardware SDR

Table 4: Pre-Lab Prediction Table for Laboratory Hardware Receiver (Table 6)

#	Parameter	Prediction	Theoretical Rationale
1	NF at Max Gain	$7.0 - 9.0$ dB	Dominated by frontend LNA noise figure as ADC noise becomes negligible.
2	ΔNF when Gain drops 20 dB	$+4.0$ to $+6.0$ dB	Lower frontend gain amplifies ADC quantization noise in Friis formula.
3	ΔP_{1dB} when Gain drops 10 dB	$+10.0$ dB	Lower gain prevents pre-ADC saturation, shifting input limit upward.
4	Y-Factor for $NF = 7$ dB, $ENR = 15$ dB	8.64 dB	$Y = 1 + \frac{ENR}{F} = 1 + \frac{31.62}{5.01} = 7.31 \implies$ 8.64 dB.
5	ΔNF with 10 dB Attenuator	$+10.0$ dB	Passive loss prior to amplification adds directly to Noise Figure.
6	ΔP_{IM3} for $+2$ dB Tone Power	$+6.0$ dB	3rd-order intermodulation products scale with a 3 dB/dB slope.
7	Is $IIP_3 - P_{\text{1dB}} \approx 9.6$ dB?	No	Hard ADC quantization clipping violates 3rd-order polynomial assumptions.
8	Team Assigned Gain Setting	Configured in Lab	Baseline parameters mapped to hardware gain setting.

7 Task 7: Risk Identification and Mitigation

1. **RF Input Overdrive Damage:** Exceeding maximum safe input power (-10 dBm) damages the SDR frontend active switches. *Mitigation:* Verify source power prior to connection, use mandatory inline attenuators, and start power sweeps from minimum levels.

2. **Receiver Saturation during Noise Source ON-State:** Excessive ENR drives the receiver into saturation during ON-state measurement, invalidating the Y-factor ratio. *Mitigation:* Turn off AGC, adjust attenuators, and verify spectral linearity in ON state.

3. **Electrostatic Discharge (ESD):** RF terminals are ESD-sensitive. *Mitigation:* Ground yourself by touching the chassis, wear ESD wristbands, and tighten coaxial connectors without excessive mechanical torque.